

Digital circuits EE1202

ASSIGNMENT_1

Deadline by Saturday 08-08-26 11:59PM

Only handwritten solving of answers I require, please stop sending code to me

1. Laboratory access controller — 4 variables

A laboratory door is controlled using four inputs:

- A : valid identity card
- B : correct PIN
- C : faculty authorization
- D : emergency mode

The door should open when:

1. The identity card and PIN are both valid.
2. The identity card is valid and faculty authorization is present.
3. Emergency mode is active and either the PIN is correct or faculty authorization is present.

Let the door-opening signal be $F(A, B, C, D)$.

Tasks:

1. Construct the truth table.
2. Express F in canonical SOP/minterm form.
3. Minimize it using a 4-variable K-map.
4. Implement the minimized expression using only two-input NAND gates.

2. Water-pump controller — 4 variables

A water pump is controlled by:

- L : water level is low
- T : storage-tank demand is active
- M : manual-start switch is active
- E : emergency shutdown is active

The pump should operate when:

1. The water level is low and the tank demands water.
2. Manual start is active and the water level is low.
3. Emergency shutdown must override every other condition.

Let the pump-control output be $P(L, T, M, E)$.

Tasks:

1. Translate the statement into a Boolean function.
2. List the output-1 minterms.
3. Minimize the function using a 4-variable K-map.
4. Realize the minimized SOP expression using only NAND gates.

3. Examination eligibility system — 5 variables

A student is permitted to write an examination based on:

- A : attendance requirement satisfied
- F : examination fee paid
- H : hall ticket available
- M : medical exemption approved
- D : disciplinary restriction active

The student is eligible when:

1. Attendance, fee payment and hall-ticket requirements are satisfied.
2. A medical exemption is approved, the fee is paid and the hall ticket is available.
3. A disciplinary restriction must deny eligibility under every condition.

Let the eligibility output be $E(A, F, H, M, D)$.

Tasks:

1. Derive the Boolean expression and truth table.
 2. Write the canonical POS expression.
 3. Minimize it using a 5-variable K-map.
 4. Implement it using only two-input NOR gates.
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4. Industrial alarm system — 5 variables

An industrial alarm has the following inputs:

- T : temperature is above its limit
- P : pressure is above its limit
- S : smoke is detected
- G : gas leakage is detected
- R : maintenance mode is active

The alarm should turn on when:

1. Any two or more of T , P and S are active.
2. Gas leakage is detected, irrespective of T , P and S .
3. During maintenance mode, temperature- and pressure-related alarms are suppressed, but smoke or gas detection must still activate the alarm.

Let the alarm output be $Y(T, P, S, G, R)$.

Tasks:

1. Construct the complete truth table.
 2. Obtain the Maxterm representation.
 3. Minimize the function using a 5-variable K-map.
 4. Implement the optimized function using only NOR gates.
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5. Smart-room lighting system — 5 variables

A room-lighting system uses:

- O : room is occupied
- D : room is dark
- M : motion is detected
- S : manual switch is ON
- E : energy-saving lock is active

The light should turn on when:

1. The room is occupied, dark and motion is detected.
2. The manual switch is ON and the room is occupied.
3. The manual switch is ON and the room is dark.
4. When energy-saving lock is active, automatic operation is disabled, but the light may still turn on if the manual switch is ON and the room is occupied.

Let the lighting output be $L(O, D, M, S, E)$.

Tasks:

1. Translate the conditions into Boolean form.
 2. List the output-1 minterms.
 3. Minimize the function using a 5-variable K-map.
 4. Design a NAND-only implementation.
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6. Machine-operation permission system — 6 variables

A machine can operate based on:

- P : main power is available
- G : safety guard is closed
- S : start button is pressed
- A : automatic mode is selected

- M : manual mode is selected
- E : emergency stop is active

The machine should operate when:

1. Power is available, the guard is closed, automatic mode is selected and the start button is pressed.
2. Power is available, the guard is closed, manual mode is selected and the start button is pressed.
3. Automatic and manual modes must not be selected simultaneously.
4. Emergency stop must disable the machine under every condition.

Let the machine output be $Y(P, G, S, A, M, E)$.

Tasks:

1. Derive the Boolean function.
2. Construct the truth table and canonical SOP expression.
3. Minimize it using a 6-variable K-map.
4. Implement the minimized function using only two-input NAND gates.

7. Secure vault controller with don't-cares — 5 variables

A secure vault uses:

- C : valid access card
- B : biometric verification passed
- K : correct security key entered
- S : supervisor authorization present
- E : emergency lockdown active

The vault opens for the following minterms:

$$F(C, B, K, S, E) = \Sigma m(12, 16, 18, 20, 22, 24, 26, 28, 30).$$

Because of hardware restrictions, the following input combinations never occur and may be treated as don't-cares:

$$d(C, B, K, S, E) = \Sigma d(13, 17, 21, 25, 29, 31).$$

Tasks:

1. Represent the function using a 5-variable K-map.
 2. Use the don't-care states to obtain the minimum SOP expression.
 3. Identify all prime implicants and essential prime implicants.
 4. Implement the minimized expression using only two-input NAND gates.
 5. Determine the total number of NAND gates required.
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8. Autonomous vehicle warning system — 6 variables

A warning system uses:

- O : obstacle detected
- L : vehicle is outside its lane
- S : speed is above the safe limit
- B : brake system has failed
- N : driving conditions are dark or foggy
- M : manual override is active

A warning must be generated when:

1. An obstacle is detected while the vehicle is speeding.
2. The vehicle leaves its lane during dark or foggy conditions.
3. Brake failure occurs with an obstacle, excessive speed or lane departure.
4. Manual override suppresses obstacle- and lane-related warnings, but it must not suppress a brake-failure warning when the vehicle is speeding.

Let the warning output be $W(O, L, S, B, N, M)$.

Tasks:

1. Develop the Boolean expression directly from the specification.
2. Generate its complete minterm list.

3. Minimize it using a 6-variable K-map.
 4. Verify that the minimized expression satisfies the manual-override requirement.
 5. Implement it using only NAND gates and calculate the NAND-gate count.
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9. Dual-tank safety controller with don't-cares — 6 variables

A chemical-processing system uses:

- A : Tank A has a high liquid level
- B : Tank B has a high liquid level
- T : temperature is high
- P : pressure is high
- V : ventilation system is operating
- M : maintenance mode is active

The emergency output Z is specified by:

$$Z(A, B, T, P, V, M) = \Sigma m(10, 11, 14, 15, 26, 27, 30, 31, 42, 43, 46, 47, 56, 57, 58, 59, 60, 61, 62, 63).$$

The following states are physically impossible and may be treated as don't-cares:

$$d(A, B, T, P, V, M) = \Sigma d(8, 9, 24, 25, 40, 41, 44, 45).$$

Tasks:

1. Plot the function on a 6-variable K-map.
 2. Find all prime implicants.
 3. Obtain the minimum POS expression using the don't-cares.
 4. Check whether more than one minimum solution exists.
 5. Implement one minimum solution using only two-input NOR gates.
 6. Calculate the number of NOR gates and maximum gate levels.
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10. Redundant power-supply controller — 6 variables

A critical system has:

- P : primary supply is healthy
- Q : secondary supply is healthy
- B : battery is healthy
- L : load demand is high
- T : system temperature is high
- M : maintenance mode is active

The system output Y should be active when:

1. At least two of the three sources P , Q and B are healthy.
2. If load demand is high, all three sources must be healthy.
3. High temperature disables normal operation.
4. During maintenance mode, the system may operate with the primary supply alone, provided load demand and temperature are both low.
5. During maintenance mode with high load demand, the normal three-source requirement still applies.

Tasks:

1. Translate the specification into an unambiguous Boolean function.
2. Write the canonical SOP expression.
3. Minimize it using a 6-variable K-map.
4. Compare the literal count before and after minimization.
5. Implement the optimized expression using only two-input NAND gates.
6. Calculate the total NAND-gate count and propagation depth.